

□ 06 Probability models example

Example

Customers, of whom 3 in 5 on average are female, arrive at a café, randomly in time, and with an arrival rate of 25 customers per hour.

The time taken to serve a customer can be modelled as normally distributed with mean 1 minute 20 seconds and standard deviation 15 seconds.

The amount spend by customers can also be modelled as normal with mean €4.35 and standard deviation €1.56 and 30% of customers pay with cash – the rest use a card.

Cups being dropped and broken is a hazard of the business and for this café, the mean time between breakages is $3\frac{1}{2}$ hours.

Calculate the likelihood of each of the following:

- (a) There will be at least two customers arriving in the next $6\frac{1}{2}$ minutes?
- (b) Two of the next six customers arriving will be male.
- (c) More than three of the next six customers arriving will be female.
- (d) Fewer than 4 of the next 9 customers will pay using cash.
- (e) The next customer will take at least a minute and a half to be served.
- (f) At least two of the next four customers will pay using a card.
- (g) The next customer will spend less than €5
- (h) At least two cups will be broken in the eight hours than the café is open.
- (i) The next customer will take 75 ± 15 seconds to be served.
- (j) Fewer than three customers will arrive in the next 10 minutes?
- (k) Two full (8 hour) days will pass with no cups being broken.

- (l) What percentage of customers spend $€4 \pm €1$?
- (m) How many customers might be expected on average in an 8-hour working day?
- (n) What customer spend would put a customer in the bottom 20%? In the top 5%?
- (o) Could the café attendant take a 5-minute break and be confident that no customers would arrive while they are gone?
- (p) How likely is it that at least 3 of the next 8 customers will spend over €5?
- (q) How likely it is that in a 6-day working week, there will be no more than one day when no customers arrive in the first 5 minutes of the café opening?